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A Leap Towards World Class Education

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Name of the Experiment: ______

Performed on: ______

Submitted on: ______

Marks	Teacher's Signature with date
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10	30/01

Q1 An algorithm is a finite sequence of well defined, & step-by-step instruction used to solved a specific problem or perform a computation.

characteristics

1. Input : An algorithm takes zero or more input
2. Output : It produces at least one output
3. Definiteness : Each step is clear, percise, unambiguous
4. Finiteness : The algorithm terminates after a finite number of step.

Q2 A data structure is a way of organization storing, and managing data in memory that it can be accessed and modified efficiently.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

An abstract Data Type (ADT) is a logical description of data type that specifies the operations that can be performed on it without specifying how these operation are implemented.

Examples.
- Data structures : Arrays
- ADT : Stack (operation : push, pop, peek)

Q3. 1. Problem Definition : Understanding requirement and constraints
2. Analysis : Identify input, output, and processing
3. Algorithm Design : Create step-by-step logic to solve the problem
4. flowchart / Pseudocode : Represent the algorithm visually or textually.
5. Coding : Convert algorithm into a program using a programming language.
6. Testing & Debugging : Check correctness & removes errors
7. Documents & maintainance : Improves & maintain the program

Role of Algorithm
the algorithm acts as a blueprint of the solution. It ensures correctness, efficiency & makes coding easier and error-free.

Q4. Linear Data Structure :
Data elements are arranged sequentially
- Array
- Stack
- Queue
- Linked List.

Non Linear Data Structure :
Data element are arranged in a hierarchical or interconnected manner

Example :- · Tree
· Graph

Q5

Algorithm	Progam
Step-by-step logic	Written code
Language indepent	Language dependent
Design Level	Implementation level
No syntax rules	Follows syntax rule.

Role in SDLC :
- Algorithm : Used in design phase to plan the solution
- Program : Used in implementation phase to execute the solution on a computer

Q6] Relationship.

- Data in raw information
- Data structure organize the data
- Algorithm process the data using data structures

Justification:
The efficiency of an algorithm depends on the data structures used

- Searching is an array → slower
- searching in a binary search tree - faster

Conclusion
Chossing the right data structure improves time and space efficiency of Algorithm